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Roll No

FT-7003-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Heat Transfer

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the basic modes of heat transfer? Explain their potential for occurrence.
b) What do you mean by critical radius of insulation? Explain its concept with help of material and surface resistance.
2. a) Explain the mechanism of convection heat transfer.
b) When is heat transfer through a fluid layer by conduction and when is it by convection? For what case, the rate of heat transfer is higher.
3. a) What is a black body? What are its properties? Explain in brief.
b) What is radiation intensity? How do you distinguish between spectral emissive power and spectral radiation intensity.
4. a) Explain the mechanism of laminar film condensation on a vertical surface.
b) What do you mean by evaporation? Discuss different types of evaporators with their applications.

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5.
 - a) What is direct contact type heat exchanger? Discuss fouling factor and its causes.
 - b) Define effectiveness of heat exchanger. How is maximum heat transfer rate obtained?
6. In a tube and tube type parallel flow heat exchanger, the hot water at 80°C is cooled to 65°C by cold water entering the heat exchanger at 20°C and leaving at 35°C . What would be the exit temperatures if the flow rates of water are doubled?
7. A horizontal plate ($k = 30 \text{ W/m. K}$) $600\text{mm} \times 30\text{mm}$ is maintained at 300°C . The air at 30°C flows over the plate. If the convection coefficient of air over the plate is $22 \text{ W/m}^2\text{K}$ and 250W heat is lost from the plate by radiation. Calculate the bottom surface temperature of the plate.
8. Write short notes on followings (any four):
 - i) Pool Boiling
 - ii) NTU
 - iii) Thermal diffusivity
 - iv) Solar radiation
 - v) Shape factor
 - vi) Laws of heat transfer
